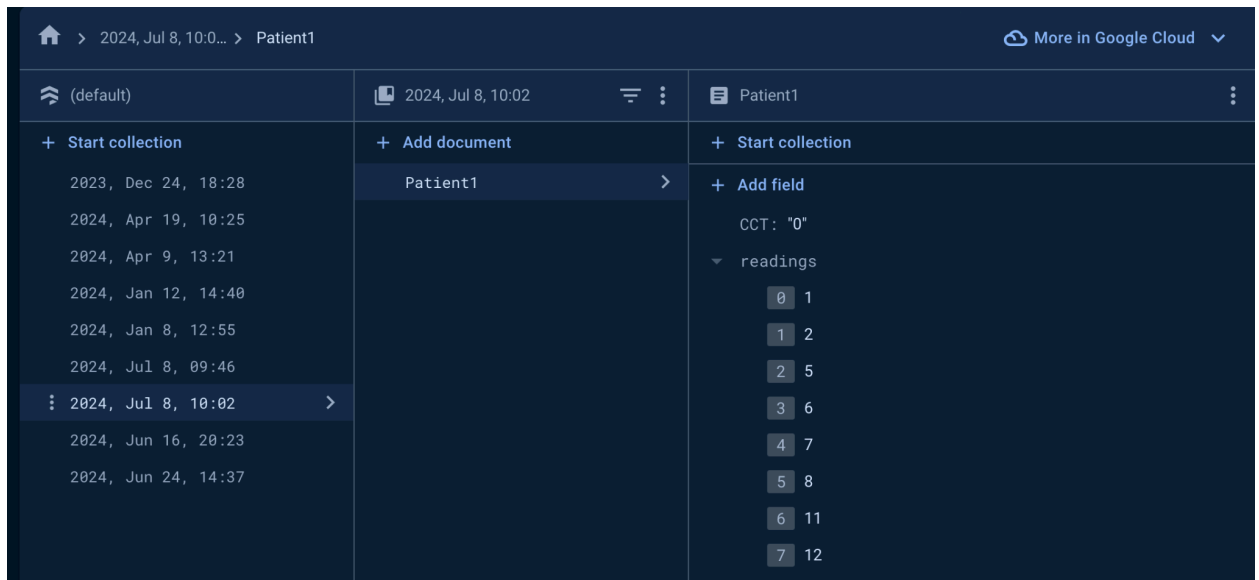


Dynamically Updating Firebase Fields

In Firebase, it can be helpful to store information using different fields to offer more information about the numbers, especially as the scope of the project increases.

1. Understanding the Current Flow

An example of how data is stored currently in Firebase is shown here:



To get this format, in `Glaucoma_App/Controllers/ConsoleViewController.swift`, the patient name, CCT, and readings values have to be specified and updated manually. As of now, they are hard coded.

```
51 // add a new specific document to the database
52 // this code chunk must be implemented before data can be written to the document in Firebase
53 db.collection(today).document("Patient1").setData([
54     "CCT": "0",
55 ])
56 print("Document successfully written!")

92 // find EXISTING document (must be defined first, otherwise values will not save, and no err
93 ref = db.collection(today).document("Patient1")
94
95 // add values to the given field
96 ref!.updateData([
97     "readings": FieldValue.arrayUnion([(notification.object! as! NSString).floatValue])
98 ])
```

The purpose of this document is to outline the application flow for dynamically updating these values during device use.

2.High-Level Flow

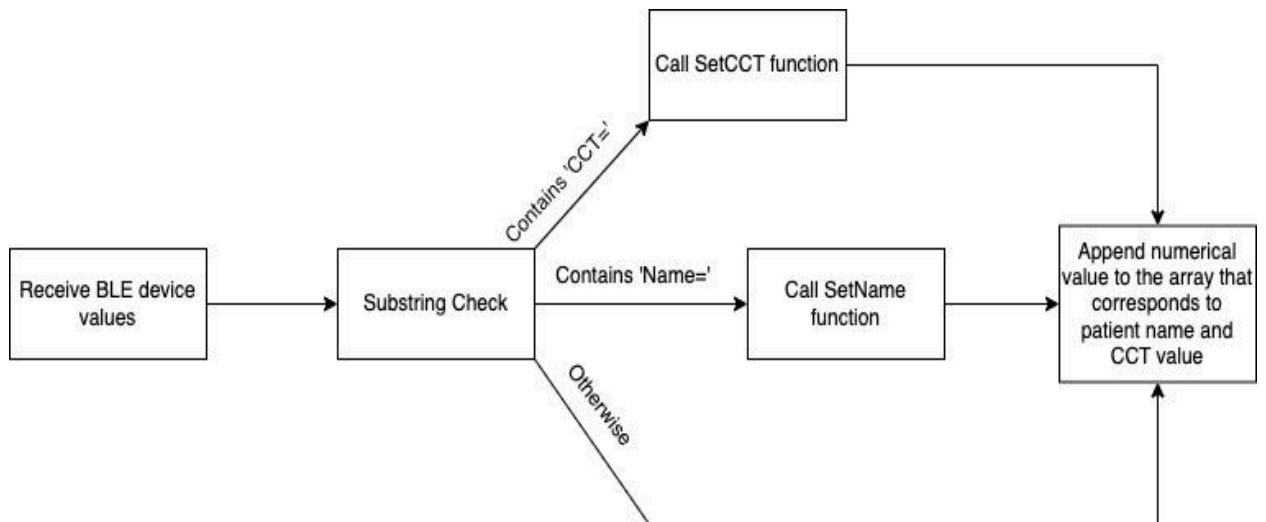
In order to support dynamic updating of the fields, it is necessary to create global variables for the patient name, CCT, and any other fields that need to be created. This will allow the application to be able to quickly update the respective field as new information comes in from the device and will still allow access to the values throughout the use of the app (thanks to persistent data!).

The file can have helper methods: setCCT, setName, etc.

If any of the fields change on the BLE device side, the device can send this to the application. For example, if the CCT needs to be updated the device can send "CCT=1". On the application side, as values come in, it is easy to check if anything starts with the string value "CCT=", and if it does, the helper method can be called to update the field.

This flow can be used for any of the fields that have the ability to dynamically change. It also can be expanded and used with any additional fields.

Below is a visual representation of the application flow.



3.Pseudo-Code

On a lower level, each box in the flow above has some additional requirements in order to be able to be updated correctly in Firebase.

In each helper method, the following lines need to be added to update the initial document in Firebase. This will create a baseline to write to - otherwise Firebase cannot

find the document and will not write to it (and will also not provide any error message so it will be difficult to diagnose).

```
// add a new specific document to the database
// this code chunk must be implemented before data can be written to the document in Firebase
db.collection(today).document(patient_name).setData([
    "CCT": cct_value,
])
print("Document successfully written!")
```